Biological control of foot rot (*Phytophthora capsici* Leonian) disease in black pepper (*Piper nigrum* L.) with rhizospheric microorganisms

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Abstract

Foot rot disease, also known as "quick wilt" caused by Phytophthora capsici Leonian in black pepper (Piper nigrum L.) often leads to mortality of the infected plants. No cultivated variety is completely resistant, though a few varieties show field tolerance. Plants in all growth stages are susceptible to the disease and the loss incurred will be huge when the plants die. Copper-based fungicides are mainly used for managing foot rot incidence in commercial plantations. Now-a-days organic black pepper has high demand in the consumer markets worldwide. Though fungicides such as copper hydroxide and copper oxychloride are permitted to be used in a restricted scale in organic agriculture, it is always advisable to avoid them as far as possible. Biological control using beneficial and antagonistic rhizospheric microorganisms is a promising alternative for controlling foot rot incidence. Both fungal and bacterial antagonists are used as biological control agents against the disease, and important among them are Trichoderma spp. and Pseudomonas fluorescens isolates. Biological control with rhizospheric microbial agents when combined with good agricultural practices, including phyto-sanitation, have shown promising results in managing the foot rot disease. Most of the rhizospheric antagonists act as plant growth promoters also. A review on the attempts made on the biological control of foot rot disease with rhizospheric microbial agents is presented here.

Introduction

Black pepper (*Piper nigrum* L.), popularly known as the "king of spices", is one of the oldest and the most popular spices in the world. It is indigenous to the Malabar

Coast of India, and its cultivation has spread to several other tropical countries like Indonesia, Malaysia, Sri Lanka, Thailand, China, Vietnam, and Cambodia (Nair, 2004). Black pepper is a plant of humid tropics requiring high rainfall and humidity. The hot and humid climate prevailing in sub mountainous tracts of Western Ghats is ideal for black pepper cultivation (Sivaraman et al., 1999; Thangaselvabal et al., 2008). Black pepper flourishes well in red laterite soil, however, it can be grown in a wide range of soils with a pH of 5.5–6.5 (IISR, 2015).

Foot rot disease caused by the soil borne oomycete pathogen *Phytophthora capsici* Leonian is the most devastating disease of black pepper that affects its production and productivity (Anandaraj and Sarma, 1995; Nair, 2004). Rhizospheric microorganisms, especially bacterial and fungal antagonists have been used as successful biocontrol agents against foot rot disease of black pepper. Selection of antagonists having potential biocontrol ability from the rhizosphere of healthy black pepper plants and screening them under *in vitro* and *in vivo* systems involves several steps before a sound recommendation is made for their use in the field conditions (Anith et al., 2002, 2003). A generalized schematic flow chart that depicts procedures involved in the selection, screening, testing for bio-efficacy and field evaluation with respect to rhizospheric bacteria is given in Fig. 1. The same methodology could be employed for selection of fungal biocontrol agents also. A review of research work done with different rhizospheric microbial biocontrol agents for suppression of *P. capsici* incidence and the strategies employed for foot rot disease management of black pepper is attempted here.

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Section snippets

Foot rot disease of black pepper

In India, the first case of *Phytophthora* disease incidence in black pepper was reported in 1902 (Barber, 1902). Isolation of *Phytophthora* sp from black pepper was reported from Karnataka state in 1929 (Rao, 1929). A further report of the disease incidence was by Samraj and Jose (1966), and *P. parasitica* var. *Piperina* was reported as the incitant. For some time, controversy existed among scientists in the taxonomic positioning of the black pepper isolate, naming it as *P. palmivora* (Holiday and

Fungal bioagents for the management of foot rot disease

The most effective fungal biocontrol agents recommended for the management of foot rot of black pepper are *Trichoderma* spp. and arbuscular mycorrhizal fungi (AMF), and they have been developed into successful commercial products. Though many other fungal biocontrol agents are tested and found effective, research works to decipher the fundamental biocontrol mechanisms of them against the foot rot pathogen is very scanty.

Bacterial bioagents for the management of foot rot disease

Several bacterial antagonists have been tested against the foot rot disease of black pepper. Jubina and Girija (1998) showed that bacterial biocontrol agents could be successfully used for the management of *P. capsici* in black pepper. Anith and Manomohandas (2001) reported the use of an *Alcaligenes* sp along with *T. harzianum* in suppressing the incidence of nursery wilt in black pepper. Though many bacterial bioagents have been tried against the foot rot disease, major success has come with the

Formulated bioagents for foot rot management

Usually carrier-based formulations, especially with talc, are used in black pepper plantations, which are applied along with organic manures to the basins of the vines. Currently, many successful products are available in the market with either organic or inorganic carrier materials. A coffee husk-based *Trichoderma* formulation has been reported to be very effective in the control of *Phytophthora* foot rot of black pepper (Sawant and Sawant, 1996). This formulation is widely being used in foot

Field level biocontrol of foot rot using formulated microbial products

For the management of foot rot of black pepper, *Trichoderma asperellum*, *Pseudomonas fluorescence* and native Arbuscular Mycorrhizal Fungi are recommended for application at the time of planting, both in the nursery and in the main field by the Kerala Agricultural University (KAU, 2016). *Trichoderma* is recommended for application around the base of the vine @ 50 g/vine (this quantity is recommended for a substrate containing *Trichoderma* @ 10^{10} cfu/g) with the onset of monsoon (May–June). A second

Conclusion and future perspectives

Foot rot disease incidence in black pepper could effectively be managed with rhizospheric microbial antagonists. However, developing bioagents for biocontrol of diseases of spice crops pose many challenges. One of the reasons is the perennial

nature of the crops. Black pepper, which takes three to four years for establishment and regular bearing is the best example for such a scenario. Biological control of foot rot disease in black pepper with microbial agents is also faced with lots of

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.